

Solvency Capital Requirement - Undertaking Specific Parameters

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1

Applications and Definitions

1.1

Unless otherwise stated, this Part applies to:

1. (1) a UK Solvency II firm; and
2. (2) in accordance with Insurance General Application 3, the Society.

1.2

In this Part, the following definitions shall apply:

accident year

means, with respect to a payment for an insurance or reinsurance claim, the year in which the insured event that gave rise to that claim took place.

adjustment factor for non-proportional reinsurance (non-life)

means the adjustment factor for non-proportional reinsurance referred to in Solvency Capital Requirement ? Standard Formula 3A4.3 and 3A4.4.

adjustment factor for non-proportional reinsurance (NSLT health)

means the adjustment factor for non-proportional reinsurance referred to in Solvency Capital Requirement ? Standard Formula 3C5.3.

aggregated losses

for the purposes of 4, means the payments made and the best estimates of the provision for insurance and reinsurance claims outstanding in segment s after the first development year of the accident year of those claims.

credibility factor

means the applicable credibility factor determined in accordance with 10.

cumulative claims amounts

means the cumulative payment amounts for insurance and reinsurance claims in segment s.

development year

means, with respect to a payment for an insurance or reinsurance claim, the difference between the year of that

payment and the accident year of that payment.

financial year

means, with respect to a payment for an insurance or reinsurance claim, the year in which this payment took place.

increase in the amount of annuity benefits (health)

means the increase in the amount of annuity benefits referred to in Solvency Capital Requirement - Standard Formula 3C15.

increase in the amount of annuity benefits (life)

means the increase in the amount of annuity benefits referred to in Solvency Capital Requirement - Standard Formula 3B5.

non-proportional reinsurance method 1

means the method set out in 8.

non-proportional reinsurance method 2

means the method set out in 9.

premium risk method

means the method set out in 4.

recognisable excess of loss reinsurance contract

1. (1) means an excess of loss reinsurance contract which:

1. (a) provides for complete compensation up to a specified limit or without limit for losses of the cedant that relate either to single insurance claims, or to all insurance claims under the same contract of insurance during a specified time period, and that are larger than a specified retention;

2. (b) covers all insurance claims that the firm may incur in the segment or homogenous risk groups within the segment during the following 12 months;

3. (c) allows for a sufficient number of reinstatements so as to ensure that all claims of multiple events incurred during the following months are covered; and

4. (d) complies with Solvency Capital Requirement - Standard Formula 3G2, 3G3, 3G5 and 3G7; and

2. (2) includes:

1. (a) arrangements with special purpose vehicles that provide risk transfer which is equivalent to that referred to in (1) to (4); and

2. (b) a combination of reinsurance contracts (which may be considered as one recognisable excess of loss reinsurance contract) where a firm has concluded several excess of loss reinsurance contracts that:

1. (i) individually meet the requirement set out in (4); and

2. (ii) in combination meet the requirements set out in (1) to (3).

recognisable stop loss reinsurance contract

1. (1) means a stop loss reinsurance contract which:

1. (a) provides for complete compensation up to a specified limit or without limit for aggregated losses of the cedant that relate to all insurance claims in the segment or homogeneous risk groups within the segment during a specified time period and that are larger than a specified retention; and

2. (b) covers all insurance claims that the firm may incur in the segment or homogenous risk groups within the segment during the following 12 months;
 3. (c) allows for a sufficient number of reinstatements so as to ensure that all claims of multiple events incurred during the following months are covered; and
 4. (d) complies with [\[/pra-rules/solvency-capital-requirement---standard-formula\]](#)[Solvency Capital Requirement - Standard Formula 3G2, 3G3, 3G5 and 3G7; and
2. (2) includes:
1. (a) arrangements with special purpose vehicles that provide risk transfer which is equivalent to that referred to in (1) to (4); and
 2. (b) a combination of reinsurance contracts (which may be considered as one recognisable stop loss reinsurance contract) where a firm has concluded several stop loss reinsurance contracts that:
 1. (i) individually meet the requirement set out in (4); and
 2. (ii) in combination meet the requirements set out in (1) to (3).

reporting year

means, with respect to a payment for an insurance or reinsurance claim, the year in which the insured event that gave rise to that claim was notified to the firm.

reserve risk method 1

means the method set out in 5.

reserve risk method 2

means the method set out in 6.

revision risk method

means the method set out in 7.

segment s

denotes the segment for which the undertaking specific parameter is determined, being a segment set out in Solvency Capital Requirement - Standard Formula 3A3 or a segment set out in Solvency Capital Requirement - Standard Formula 3C4, as specified in the firm's USP permission.

standard deviation for non-life gross premium risk

means the standard deviation for non-life gross premium risk referred to in Solvency Capital Requirement ? Standard Formula 3A4.3.

standard deviation for non-life premium risk

means the standard deviation for non-life premium risk referred to in Solvency Capital Requirement - Standard Formula 3A4.2(1).

standard deviation for non-life reserve risk

means the standard deviation for non-life reserve risk referred to in Solvency Capital Requirement - Standard Formula 3A4.2(2).

standard deviation for NSLT health gross premium risk

means the standard deviation for NSLT health gross premium risk referred to in Solvency Capital Requirement - Standard Formula 3C5.3.

standard deviation for NSLT health premium risk

means the standard deviation for NSLT health premium risk referred to in Solvency Capital Requirement - Standard Formula 3C5.2(1).

standard deviation for NSLT health reserve risk

means the standard deviation for NSLT health reserve risk referred to in Solvency Capital Requirement - Standard Formula 3C5.2(2).

2

Undertaking Specific Parameters

2.1

A firm must not apply an undertaking specific parameter unless it is a USP firm.

2.2

A USP firm must not revert to using the standard parameter in respect of which it has been granted a USP Permission.

2.3

A USP firm must only use a USP method that corresponds to the standard parameter in respect of which it has been granted a USP Permission, as determined in accordance with the following table:

Standard parameter in respect of which the firm has been granted a USP Permission	Corresponding applicable USP method
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in the non-life premium and reserve risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3A1, for each segment set out in Solvency Capital Requirement ? Standard Formula 3A3	the standard deviation for non-life premium risk premium risk method
	the standard deviation for non-life gross premium risk premium risk method
	the adjustment factor for non-proportional reinsurance (non-life)
1. (1) where there is a recognisable excess of loss reinsurance contract, non-proportional reinsurance method 1; and	
2. (2) where there is a recognisable stop loss reinsurance contract, non-proportional reinsurance method 2	
the standard deviation for non-life reserve risk reserve risk method 1 or reserve risk method 2	
in the life revision risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3B5	the increase in the amount of annuity benefits (life) revision risk method
in the NSLT health premium and reserve risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3C2, for each segment set out in Solvency Capital Requirement ? Standard Formula 3C4	the standard deviation for NSLT health premium risk premium risk method
	the standard deviation for NSLT health gross premium risk premium risk method
	the adjustment factor for non-proportional reinsurance (NSLT health)
1. (1) where there is a recognisable excess of loss reinsurance contract, non-proportional reinsurance method 1; and	

in the non-life premium and reserve risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3A1, for each segment set out in Solvency Capital Requirement ? Standard Formula 3A3

the standard deviation for non-life premium risk | premium risk method

the standard deviation for non-life gross premium risk | premium risk method

the adjustment factor for non-proportional reinsurance (non-life) |

1. (1) where there is a recognisable excess of loss reinsurance contract, non-proportional reinsurance method 1; and

2. (2) where there is a recognisable stop loss reinsurance contract, non-proportional reinsurance method 2

the standard deviation for non-life reserve risk | reserve risk method 1 or reserve risk method 2

in the life revision risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3B5

the increase in the amount of annuity benefits (life) | revision risk method

in the NSLT health premium and reserve risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3C2, for each segment set out in Solvency Capital Requirement ? Standard Formula 3C4

the standard deviation for NSLT health premium risk | premium risk method

the standard deviation for NSLT health gross premium risk | premium risk method

the adjustment factor for non-proportional reinsurance (NSLT health) |

1. (1) where there is a recognisable excess of loss reinsurance contract, non-proportional reinsurance method 1; and

2. (2) where there is a recognisable stop loss reinsurance contract, non-proportional reinsurance method 2

the standard deviation for NSLT health reserve risk

| reserve risk method 1 or reserve risk method 2

in the health revision risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3C15

the __increase in the amount of annuity benefits (health) | revision risk method

2.4

Where, in accordance with 2.3, a USP firm is permitted to select from alternative USP methods:

1. (1) the firm must use the USP method that produces the most accurate result for the purposes of fulfilling the calibration requirements in Solvency Capital Requirement ? General Provisions 3.3 and 3.4; and

2. (2) the firm must use the USP method that produces the most conservative result where it is not able to demonstrate the greater accuracy of the results of one USP method over another USP method.

2.5

1. (1) For each segment set out in Solvency Capital Requirement ? Standard Formula 3C5.2(2), a firm must not replace both of the following standard parameters:

1. (a) the standard deviation for non-life gross premium risk; and

2. (b) the adjustment factor for non-proportional reinsurance (non-life).

2. (2) For each segment set out in Solvency Capital Requirement ? Standard Formula 3C4, a firm must not replace both of the following standard parameters:

(a) the standard deviation for NSLT health gross premium risk; and

(b) the adjustment factor for non-proportional reinsurance (health).

3

Data Criteria

3.1

A USP firm must ensure that data used to calculate an undertaking specific parameter is complete, accurate, and appropriate.

3.2

For the purposes of 3.1, a firm must not treat data as complete, accurate, and appropriate unless they satisfy all the following criteria:

1. (1) the data meet the conditions set out in Technical Provisions - Further Requirements 4(1), 4(2) and 4(3), and the firm complies in relation to that data with the requirements set out in Technical Provisions - Further Requirements 4(4), where any reference to the calculation of ?technical provisions? is to be interpreted for these purposes as a reference to the calculation of an ?undertaking specific parameter?;

2. (2) the data are capable of being incorporated into the USP method;

3. (3) the data do not prevent the firm from complying with the requirements of Solvency Capital Requirement ? General Provisions 3.3 and 3.4;

4. (4) the data meet any additional USP method-specific data requirements, as set out for each USP method; and

5. (5) the data and its production process are thoroughly documented, including:

1. (a) the collection of data and analysis of its quality, where the documentation required includes a directory of the data, specifying its source, characteristics, and usage and the specification for the collection, processing and application of the data;
2. (b) the choice of assumptions used in the production and adjustment of the data, including adjustments with regard to reinsurance and catastrophe claims and about the allocation of expenses, where the documentation required includes a directory of all relevant assumptions that the calculation of technical provisions is based upon and a justification for the choice of the assumptions;
3. (c) the selection and application of actuarial and statistical methods for the production and the adjustment of the data; and
4. (d) the validation of the data.

3.3

Where external data are used, a USP firm must also ensure those data satisfy all of the following additional criteria:

1. (1) the process for collecting the data is transparent, auditable and known by the firm that uses the data to calculate the undertaking specific parameter;
2. (2) where the data stem from different sources, the assumptions made in the collection, processing and application of data ensure that the data are comparable;
3. (3) the data stem from firms for which the business nature and risk profiles are similar to that of the firm that uses that data to calculate the undertaking specific parameter;
4. (4) the firm that uses that data to calculate the undertaking specific parameter is able to verify that there is sufficient statistical evidence that the probability distributions underlying its own data and the external data have a high degree of similarity, in particular with respect to the level of volatility they reflect;
5. (5) external data only comprise data from firms with a similar risk profile;
6. (6) the risk profile referred to in (5) is similar to the risk profile of the firm that uses the data to calculate the undertaking specific parameter; and
7. (7) for the purposes of (5) and (6), when considering whether the risk profiles are similar, a firm must consider in particular whether the external data comprise data from firms for which the business nature and risk profiles with respect to the external data are similar and for which there is sufficient statistical evidence that the probability distributions underlying the external data exhibit a high degree of homogeneity.

4

Premium Risk Method

Input data and USP method-specific data requirements

4.1

A USP firm using the premium risk method to calculate an undertaking specific parameter must ensure that the data for estimating the standard deviation of segment s only consist of the following:

1. (1) aggregated losses; and
2. (2) the earned premiums in segment s .

4.2

The aggregated losses and earned premiums referred to in 4.1 must be available separately for each accident year of the insurance and reinsurance claims in segment s .

4.3

A USP firm using the premium risk method must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the premium risk that the firm is exposed to during the following 12 months;
2. (2) data are available for at least five consecutive accident years;
3. (3) where the premium risk method is applied to replace the standard parameters for either the standard deviation for non-life gross premium risk or the standard deviation for NSLT health gross premium risk, the aggregated losses and earned premiums are not adjusted for recoverables from reinsurance contracts and special purpose vehicles or premiums from reinsurance contracts;
4. (4) where the premium risk method is applied to replace the standard parameters for either the standard deviation for non-life premium risk or the standard deviation for NSLT health premium risk:
 1. (a) the aggregated losses are adjusted for amounts recoverable from reinsurance contracts and special purpose vehicles which are consistent with the reinsurance contracts and special purpose vehicles that are in place to provide cover for the following 12 months; and
 2. (b) the earned premiums are adjusted for premiums from reinsurance contracts which are consistent with the reinsurance contracts and special purpose vehicles that are in place to provide cover for the following 12 months;
5. (5) the aggregated losses are adjusted for catastrophe claims to the extent that the risk of those claims is reflected in the non-life catastrophe risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3A7 or the health catastrophe risk sub-module referred to in Solvency Capital Requirement ? Standard Formula 3C17;
6. (6) the aggregated losses include the expenses incurred in servicing the insurance and reinsurance obligations; and
7. (7) the data are consistent with the following assumptions:
 1. (a) expected aggregated losses in a particular segment and accident year are linearly proportional in earned premiums in a particular accident year;
 2. (b) the variance of aggregated losses in a particular segment and accident year is quadratic in earned premiums in a particular accident year;
 3. (c) aggregated losses follow a lognormal distribution; and
 4. (d) maximum likelihood estimation is appropriate.

USP method specification

4.4

For the purposes of 4.5 to 4.8, the following notations apply:

1. (1) accident years are denoted by consecutive numbers starting with 1 for the first accident year for which data are available;
2. (2) T denotes the latest accident year for which data are available;
3. (3) for all accident years, the aggregated losses in segment s in a particular accident year t are denoted by y_{st} ; and
4. (4) for all accident years, the earned premiums in segment s in a particular accident year t are denoted by x_{st} .

4.5

A USP firm using the premium risk method must calculate the standard deviation of segment s in accordance with the following formula:

$$\sigma_{(prem,s,USP)} = c \cdot \hat{\sigma} \left(\widehat{\delta}, \hat{\gamma} \right) \cdot \sqrt{\frac{T+1}{T-1}} + \left(1 - c \right) \cdot \sigma_{(prem,s)}$$

- (1) c denotes the credibility factor;

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- (2) $\hat{\sigma}$ denotes the standard deviation function set out in [4.6];
- (3) $\hat{\delta}$ denotes the mixing parameter set out in [4.7];
- (4) $\hat{\gamma}$ denotes the logarithmic variation coefficient set out in [4.7]; and
- (5) $\sigma_{(prem,s)}$ denotes the standard parameter that is replaced by the firm's undertaking specific parameter in respect of segment s .

4.6

The standard deviation function must be equal to the following function of two variables:

$$\hat{\sigma}(\hat{\delta}, \hat{\gamma}) = \exp\left(\hat{\gamma} + \frac{1}{2} \sum_{t=1}^T \pi_t(\hat{\delta}, \hat{\gamma}) \cdot \ln\left(\frac{y_t}{x_t}\right)\right) \sum_{t=1}^T \pi_t(\hat{\delta}, \hat{\gamma})$$

where:

(1) $\hat{\delta}$ and $\hat{\gamma}$ are defined in [4.5(3) and (4), respectively;

(2) $\exp$ denotes the exponential function;

(3) $\ln$ denotes the natural logarithm function; and

(4) π_t denotes the following function of two variables:

$$\pi_t(\hat{\delta}, \hat{\gamma}) = \frac{1}{\ln\left(1 + \left((1 - \hat{\delta}) \cdot \frac{\bar{x}}{x_t} + \hat{\delta}\right) \cdot e^{2 \cdot \hat{\gamma}}\right)}$$

where:

(a) $\hat{\delta}$ and $\hat{\gamma}$ are defined in [4.5(3) and (4), respectively; and

(b) $\bar{x}$ denotes the following amount:

$$\bar{x} = \frac{1}{T} \sum_{t=1}^T x_t$$

4.7

The mixing parameter and the logarithmic variation coefficient must be the values $\hat{\delta}$ and $\hat{\gamma}$ respectively for which the following amount becomes minimal:

$$\sum_{t=1}^T \pi_t(\hat{\delta}, \hat{\gamma}) \left(\ln\left(\frac{y_t}{x_t}\right) + \frac{1}{2} \pi_t(\hat{\delta}, \hat{\gamma}) + \hat{\gamma} - \ln\left(\hat{\sigma}(\hat{\delta}, \hat{\gamma})\right) \right)^2 - \sum_{t=1}^T \ln\left(\pi_t(\hat{\delta}, \hat{\gamma})\right)$$

where:

(1) $\ln$ denotes the natural logarithm function;

(2) π_t denotes the function set out in [4.6(4);

(3) $\hat{\sigma}$ denotes the standard deviation function set out in [4.6; and

(4) $\bar{x}$ denotes the following amount:

$$\bar{x} = \frac{1}{T} \sum_{t=1}^T x_t$$

4.8

For the purposes of 4.7, a USP firm must ensure that no values for the mixing parameter less than zero or exceeding 1 are considered for the determination of the minimal amount.

5

Reserve Risk Method 1

Input data and USP method-specific data requirements

5.1

A USP firm using reserve risk method 1 to calculate an undertaking specific parameter must ensure that the data for estimating the standard deviation of segment s consist of the following:

1. (1) the sum of the best estimate provision at the end of the financial year for claims that were outstanding in segment s at the beginning of the financial year and the payments made during the financial year for claims that were outstanding in segment s at the beginning of the financial year; and
2. (2) the best estimate of the provision for claims outstanding in segment s at the beginning of the financial year.

5.2

The amounts referred to in 5.1(1) and 5.1(2) must be available separately for different financial years.

5.3

A USP firm using reserve risk method 1 must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the reserve risk that the firm is exposed to during the following 12 months;
2. (2) data are available for at least five consecutive financial years;
3. (3) the data are adjusted for amounts recoverable from reinsurance contracts and special purpose vehicles which are consistent with the reinsurance contracts and special purpose vehicles that are in place to provide cover for the following 12 months;
4. (4) the data includes the expenses incurred in servicing the insurance and reinsurance obligations; and
5. (5) the data are consistent with the following assumptions:
 1. (a) the amount referred to in 5.1(1) in that particular segment and financial year is linearly proportional in the best estimate of the provision for claims outstanding in that particular segment and financial year;
 2. (b) the variance of the amount referred to 5.1(1) in a particular segment and financial year is quadratic in the provision for claims outstanding in a particular segment and financial year;
 3. (c) the amount referred to in 5.1(1) follows a lognormal distribution; and
 4. (d) maximum likelihood estimation is appropriate.

USP method specification

5.4

For the purposes of 5.5 to 5.8, the following notations apply:

1. (1) the financial years are denoted by consecutive numbers starting with 1 for the first financial year for which data are available;
2. (2) $_T$ denotes the latest financial year for which data are available;
3. (3) for all financial years, the amount referred to 5.1(1) in segment s in a particular financial year $_t$ is denoted by $_{y_t}$; and
4. (4) for all financial years, the best estimate of the provision for claims outstanding in segment s in a particular financial year $_t$ is denoted by $_{x_t}$.

5.5

A USP firm using reserve risk method 1 must calculate the standard deviation of segment s in accordance with the following formula:

$$\hat{\sigma}_{\left(\text{res}, s, \text{USP}\right)} = c \cdot \hat{\sigma}_{\left(\hat{\delta}, \hat{\gamma}\right)} \cdot \sqrt{\frac{T+1}{T-1}} + \sqrt{(1-c) \cdot \sigma_{\left(\text{res}, s\right)}}$$

where:

- (1) | c denotes the credibility factor;
- (2) | $\hat{\sigma}$ denotes the standard deviation function set out in [5.6];
- (3) | $\hat{\delta}$ denotes the mixing parameter set out in [5.7];
- (4) | $\hat{\gamma}$ denotes the logarithmic variation coefficient set out in [5.7]; and
- (5) | $\sigma_{\left(\text{res}, s\right)}$ denotes the standard parameter that is replaced by the firm's undertaking specific parameter σ in respect of segment s .

5.6

The standard deviation function must be equal to the following function of two variables:

$$\hat{\sigma}_{\left(\hat{\delta}, \hat{\gamma}\right)} = \exp\left(\hat{\gamma} + \frac{1}{2} \sum_{t=1}^T \pi_t \left(\hat{\delta}, \hat{\gamma}\right) \cdot \ln\left(\frac{y_t}{x_t}\right)\right) \cdot \sum_{t=1}^T \pi_t \left(\hat{\delta}, \hat{\gamma}\right)$$

where:

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- (1) | $\hat{\delta}$ and $\hat{\gamma}$ are defined in [5.5(3) and 5.5(4), respectively;
- (2) | $\exp$ denotes the exponential function;
- (3) | $\ln$ denotes the natural logarithm function; and
- (4) | π_t denotes the following function of two variables:

$$\pi_t \left(\hat{\delta}, \hat{\gamma}\right) = \frac{1}{\ln\left(1 + \left(1 - \hat{\delta}\right) \cdot \frac{\bar{x}}{x_t} + \hat{\delta}\right)} \cdot e^{2 \cdot \hat{\gamma}}$$

| where:

- (a) | $\hat{\delta}$ and $\hat{\gamma}$ are defined in [5.5(3) and 5.5(4), respectively; and
- (b) | $\bar{x}$ denotes the following amount:
 $\bar{x} = \frac{1}{T} \cdot \sum_{t=1}^T x_t$

5.7

The mixing parameter and the logarithmic variation coefficient must be the values $\hat{\delta}$ and $\hat{\gamma}$ respectively for which the following amount becomes minimal:

$$\sum_{t=1}^T \pi_t \left(\hat{\delta}, \hat{\gamma}\right) \cdot \left(\ln\left(\frac{y_t}{x_t}\right) + \frac{1}{2} \cdot \pi_t \left(\hat{\delta}, \hat{\gamma}\right) + \hat{\gamma} - \ln\left(\hat{\sigma}_{\left(\hat{\delta}, \hat{\gamma}\right)}\right)\right)^2 - \sum_{t=1}^T \ln\left(\pi_t \left(\hat{\delta}, \hat{\gamma}\right)\right)$$

where:

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- (1) $\ln$ denotes the natural logarithm function;
- (2) π_t denotes the function set out in [5.6(4)];
- (3) $\hat{\sigma}$ denotes the standard deviation function set out in [5.6]; and
- (4) $\bar{x}$ denotes the following amount:
- $$\bar{x} = \frac{1}{T} \cdot \sum_{t=1}^T x_t$$

5.8

For the purposes of 5.7, a USP firm must ensure that no values for the mixing parameter less than zero or exceeding 1 are considered for the determination of the minimal amount.

6

Reserve Risk Method 2

Input data and USP method-specific data requirements

6.1

A USP firm using reserve risk method 2 to calculate an undertaking specific parameter must ensure that the data for estimating the standard deviation of segment s consist of cumulative claims amounts, separately for each accident year and development year of the payments.

6.2

A USP firm using reserve risk method 2 must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the reserve risk that the firm is exposed to during the following 12 months;
2. (2) data are available for at least five consecutive accident years;
3. (3) in the first accident year, data are available for at least five consecutive development years;
4. (4) in the first accident year the cumulative claims amounts of the latest development year for which data are available includes all the payments of the accident year except an immaterial amount;
5. (5) the number of consecutive accident years for which data are available is not less than the number of consecutive development years in the first accident year for which data are available;
6. (6) the cumulative claims amounts are adjusted for amounts recoverable from reinsurance contracts and special purpose vehicles which are consistent with the reinsurance contracts and special purpose vehicles that are in place to provide cover for the following 12 months;
7. (7) the cumulative claims amounts must include the expenses incurred in servicing the insurance or reinsurance obligations; and
8. (8) the data are consistent with the following assumptions about the stochastic nature of cumulative claims amounts:
 1. (a) cumulative claims amounts for different accident years are mutually stochastically independent;
 2. (b) for all accident years the implied incremental claim amounts are stochastically independent;
 3. (c) for all accident years the expected value of the cumulative claims amount for a development year is proportional to the cumulative claims amount for the preceding development year; and
 4. (d) for all accident years the variance of the cumulative claims amount for a development year is proportional to the cumulative claims amount for the preceding development year.

6.3

Page 12/21

$$| \sum_{j=0}^{I-j-1} C_{(i,j)} |$$

(3) $| S_j |$ denotes for a specific development year j the following amount:

$$| \sum_{j=0}^{I-j-1} C_{(i,j)} |$$

(4) $| \hat{Q}_j |$ denotes for a specific development year j the following amount:

$$| \hat{Q}_j = \frac{\hat{\sigma}_j^2}{\hat{f}_j^2} |$$

where:

(a) $| \hat{f}_j |$ denotes the development factor estimate of development year j as set out in [6.5(3); and

(b) $| \hat{\sigma}_j^2 |$ denotes the following amount:

$$| \hat{\sigma}_j^2 = \frac{1}{I-j-1} \sum_{i=0}^{I-j-1} C_{(i,j)} \left(\frac{C_{(i,j+1)}}{C_{(i,j)}} - \hat{f}_j \right)^2 | \text{ where } j = 0, \dots, (J-2)$$

$$| \hat{\sigma}_j^2 = \min \left(\hat{\sigma}_{J-2}^2, \hat{\sigma}_{J-3}^2 \frac{\hat{\sigma}_{J-2}^4}{\hat{\sigma}_{J-3}^2} \right) | \text{ where } j = (J-1)$$

7

Revision Risk Method

7.1

1. (1) A USP firm must only use the revision risk method for:

1. (a) the life revision risk sub-module referred to in Solvency Capital Requirement - Standard Formula 3B5; or
2. (b) the health revision risk sub-module referred to in Solvency Capital Requirement - Standard Formula 3C15,
3. if the annuities within scope of the relevant sub-module are not subject to material inflation risk.

2. (2) For the purposes of (1), a firm must treat inflation risk as material where ignoring it in the calculation of the sub-modules referred to in (1)(a) and (1)(b) could influence the decision-making or the judgment of the users of the information, including the supervisory authorities.

Input data and USP method-specific data requirements

7.2

A USP firm using the revision risk method to calculate an undertaking specific parameter must ensure that the data for estimating the increase in the amount of annuity benefits consist of annual amounts of annuity benefits of annuity insurance obligations where the benefits payable could increase as a result of changes in the legal environment or in the state of health of the person insured, separately for consecutive financial years and each beneficiary.

7.3

A USP firm using the revision risk method must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the revision risk that the firm is exposed to during the following 12 months;
2. (2) data are available for at least five consecutive financial years;
3. (3) the annuity benefits are gross, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;
4. (4) the annuity benefits must include the expenses incurred in servicing the annuity obligations; and
5. (5) the data are consistent with the following assumptions about the stochastic nature of increases in the amount of annuity benefits:
 1. (a) the annual number of increases in annuity benefits follows a negative binomial distribution, including in the tail of

the distribution;

2. (b) the amount of an increase in annuity benefits follows a lognormal distribution, including in the tail of the distribution; and

3. (c) the annual number of increases in annuity benefits and the amounts of the increases in annuity benefits are mutually stochastically independent.

USP method specification

7.4

For the purposes of 7.5 to 7.9, the following notations apply:

(1) | the financial years are denoted by consecutive numbers starting with 1 for the first financial year for which data are available;

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(2) | T denotes the latest financial year for which data are available;

(3) | $A(i, t)$ denotes the annuity benefits of beneficiary i in financial year t ; and

(4) | $D(i, t)$ denotes the change of annuity benefits after financial year t , being equal to the following difference:

$$D(i, t) = A(i, t) - A(i, t-1)$$

7.5

A USP firm using reserve risk method --- must calculate the increase in the amount of annuity benefits in accordance with the following formula:

$$S_{\text{USP}} = c \cdot \frac{\text{VaR}_{0.995}(R) - \bar{R}}{\bar{R}} + (1-c) \cdot S$$

where: |

---|---

(1) | c denotes the credibility factor;

(2) | $\bar{R}$ denotes the expected value of increases in annuity benefits set out in [7.6];

(3) | $\text{VaR}_{0.995}(R)$ denotes the 99.5% quantile of the distribution of increases in annuity benefits set out in [7.7];

(4) | S is:

| (a) | equal to 3% where the calculation relates to the life revision risk sub-module referred to in Solvency Capital Requirement - Standard Formula 3B5; and

| (b) | equal to 4% where the calculation relates to the health revision risk sub-module referred to in Solvency Capital Requirement - Standard Formula 3C15.

7.6

The expected value of increases in annuity benefits must be equal to the following:

$$\bar{R} = \bar{X} \cdot \bar{N}$$

where: |

---|---

(1) | $\bar{X}$ denotes the estimated average change in annuity benefits, restricted to those changes in annuity benefits that are larger than zero; and

(2) | $\bar{N}$ denotes the estimated average number, per financial year, of changes in annuity benefits that are larger

Input data and USP method-specific data requirements

8.2

A USP firm using non-proportional reinsurance method 1 to calculate an undertaking specific parameter μ must ensure that the data for estimating an adjustment factor for non-proportional reinsurance consist of the ultimate claim amounts of insurance and μ reinsurance μ claims μ that were reported to the firm in segment s during the preceding financial years, separately for each insurance and reinsurance μ claim.

8.3

A USP firm using non-proportional reinsurance method 1 must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the premium risk that the firm is exposed to during the following 12 months;
2. (2) the data do not indicate a higher premium risk than reflected in the corresponding standard deviation for premium risk used to calculate the SCR using the standard formula;
3. (3) the ultimate claim amounts are estimated in the year the insurance and reinsurance claims were reported;
4. (4) data are available for at least five reporting years;
5. (5) where the recognisable excess of loss reinsurance contract applies to gross claims, the ultimate claim amounts are gross;
6. (6) where the recognisable excess of loss reinsurance contract applies to claims after deduction of the recoverables from certain other reinsurance contracts and special purpose vehicles, the amounts receivable from those reinsurance contracts and special purpose vehicles are deducted from the ultimate claim amounts;
7. (7) the ultimate claim amounts must not include expenses incurred in servicing the insurance and reinsurance obligations; and
8. (8) the data are consistent with the assumption that ultimate claim amounts follow a lognormal distribution, including in the tail of the distribution.

USP method specification

8.4

For the purposes of 8.5 to 8.8, the following notations apply:

1. (1) insurance and reinsurance claims for which data are available are denoted by consecutive numbers starting with 1;
2. (2) n denotes the number of insurance and reinsurance claims for which data are available;
3. (3) Y_i denotes the ultimate claim amount of the insurance and reinsurance claims i ;
4. (4) μ and σ^2 denote the first and second moment, respectively, of the claim amount distribution, being equal to the following amounts:

$$\mu = \frac{1}{n} \sum_{i=1}^n Y_i$$

and

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n Y_i^2$$

1. (5) b_1 denotes the amount of the retention of the recognisable excess of loss reinsurance contract;
2. (6) where the recognisable excess of loss reinsurance contract provides compensation only up to a specified limit, b_2 denotes the amount of that limit.

8.5

A USP firm __ using __non-proportional reinsurance method 1 __**_ must calculate the adjustment factor for non-proportional reinsurance in accordance with the following formula:

$$\mathbb{N}[\{NP\}_{USP}] = c \cdot NP^{\prime} + (1 - c) \cdot NP$$

where:

1. (1) c denotes the credibility factor;
2. (2) NP denotes the estimated adjustment factor for non-proportional reinsurance set out in 8.6; and
3. (3) NP denotes the standard parameter that is replaced by the firm's undertaking specific parameter.

8.6

The estimated adjustment factor for non-proportional reinsurance must be equal to the following:

$$NP = \sqrt{\frac{\omega_1 - \omega_2 + \omega + 2 \cdot (b_2 - b_1) \cdot (\mu_2 - \mu)}{\omega}} \quad \text{where [8.4(6) applies,}$$

---|---|---

$$\sqrt{\frac{\omega_1}{\omega}} \quad \text{otherwise.}$$

where the parameters ω , ω_1 and ω_2 are set out in 8.7.

8.7

The parameters μ_2 , μ_1 and μ must be equal to the following, respectively:

$$\mu_2 = \mu \cdot N\left(\frac{\ln(b_2 - \theta)}{\eta} - \eta\right) + b_2 \cdot N\left(-\frac{\ln(b_2 - \theta)}{\eta}\right)$$

$$\mu_1 = \mu \cdot N\left(\frac{\ln(b_1 - \theta)}{\eta} - 2\eta\right) + b_1^2 \cdot N\left(-\frac{\ln(b_1 - \theta)}{\eta}\right)$$

$$\mu = \mu \cdot N\left(\frac{\ln(b_2 - \theta)}{\eta} - 2\eta\right) + b_2^2 \cdot N\left(-\frac{\ln(b_2 - \theta)}{\eta}\right)$$

where:

1. (1) N denotes the cumulative distribution function of the normal distribution;
2. (2) $\ln$ denotes the natural logarithm function; and
3. (3) the parameters θ and η are equal to the following, respectively:

$$\theta = 2 \ln \mu - \frac{1}{2} \ln \omega$$

and

$$\eta = \sqrt{\ln \omega - 2 \ln \mu}$$

8.8

Notwithstanding 8.6, where non-proportional reinsurance covers homogeneous risk groups within a segment, the estimated adjustment factor for non-proportional reinsurance must be equal to the following:

$$NP = \frac{\sum_h V(\text{prem}, h) \cdot NP_h}{\sum_h V(\text{prem}, h)}$$

where:

1. (1) $V_{(prem,h)}$ denotes the volume measure for premium risk of the homogeneous risk group h determined in accordance with Solvency Capital Requirement - Standard Formula 3A2.3; and
2. (2) $NP(h)$ denotes the estimated adjustment factor for non-proportional reinsurance of homogeneous risk group h determined in accordance with 8.6.

9

Non-Proportional Reinsurance Method 2

9.1

A USP firm must only use non-proportional reinsurance method 2 if there is a recognisable stop loss reinsurance contract in place.

Input data and USP method-specific data requirements

9.2

A USP firm using the non-proportional reinsurance method 2 to calculate an undertaking specific parameter must ensure that data for estimating an adjustment factor for non-proportional reinsurance consist of the aggregated annual losses of insurance and reinsurance claims that were reported to the firm in segment s during the preceding financial years.

9.3

A USP firm using the non-proportional reinsurance method 2 must ensure that the data used to calculate an undertaking specific parameter satisfy all the following USP method-specific data requirements:

1. (1) the data are representative of the premium risk that the firm is exposed to during the following 12 months;
2. (2) the data do not indicate a higher premium risk than reflected in the corresponding standard deviation for premium risk used to calculate the SCR using the standard formula;
3. (3) the aggregated annual losses are estimated in the year insurance and reinsurance claims were reported;
4. (4) data are available for at least five reporting years;
5. (5) where the recognisable stop loss reinsurance contract applies to gross claims, the aggregated annual losses are gross;
6. (6) where the recognisable stop loss reinsurance contract applies to claims after deduction of the recoverables from certain other reinsurance contracts and special purpose vehicles, the amounts receivable from those reinsurance contracts and special purpose vehicles are deducted from the aggregated annual losses;
7. (7) the aggregated annual losses must not include expenses incurred in servicing the insurance and reinsurance obligations; and
8. (8) the data are consistent with the assumption that aggregated annual losses follow a lognormal distribution, including in the tail of the distribution.

USP method specification

9.4

For the purposes of 9.5 to 9.8, the following notations apply:

1. (1) n denotes the number of financial years for which aggregated annual losses data are available;
2. (2) Y_i denotes the aggregated losses in financial year i ;
3. (3) μ and σ denote the first and second moment, respectively, of the aggregated annual losses distribution, being equal to the following amounts:

$$\mu = \frac{1}{n} \sum_{i=1}^n Y_i$$

and

$$\omega = \frac{1}{n} \sum_{i=1}^n Y_i^2$$

- (4) b_1 denotes the amount of the retention of the recognisable stop loss reinsurance contract; and
- (5) where the recognisable stop loss reinsurance contract provides compensation only up to a specified limit, b_2 denotes the amount of that limit.

9.5

A USP firm using the non-proportional reinsurance method 2 must calculate the adjustment factor for non-proportional reinsurance in accordance with the following formula:

$$\{NP\}_{USP} = c \cdot NP^{\prime} + (1-c) \cdot NP$$

where:

- (1) c denotes the credibility factor;
- (2) $NP^{\prime}$ denotes the estimated adjustment factor for non-proportional reinsurance set out in 9.6; and
- (3) NP denotes the standard parameter that is replaced by the firm's undertaking specific parameter.

9.6

The estimated adjustment factor for non-proportional reinsurance must be equal to the following:

$$\sqrt{\frac{\left(\omega_1 + \omega - \omega_2 + 2\left(b_2 - b_1\right)\left(\mu_2 - \mu\right)\right) - \left(\mu_1 + \mu - \mu_2\right)^2}{\omega - \mu^2}} \quad \text{where [9.4(5) applies,} \\ \text{---|---|---} \\ \sqrt{\frac{\omega - \mu_1^2}{\omega - \mu^2}} \quad \text{otherwise.}$$

where the parameters μ_1 , μ_2 , b_1 and b_2 are set out in 9.7.

9.7

The parameters μ_1 , μ_2 , b_1 and b_2 must be equal to the following, respectively:

$$\mu_1 = \mu \cdot N\left(\frac{\ln(b_1 - \theta)}{\eta} - \eta\right) + b_1 \cdot N\left(-\frac{\ln(b_1 - \theta)}{\eta}\right)$$

$$\mu_2 = \mu \cdot N\left(\frac{\ln(b_2 - \theta)}{\eta} - \eta\right) + b_2 \cdot N\left(-\frac{\ln(b_2 - \theta)}{\eta}\right)$$

$$\omega_1 = \omega \cdot N\left(\frac{\ln(b_1 - \theta)}{\eta} - 2\eta\right) + b_1^2 \cdot N\left(-\frac{\ln(b_1 - \theta)}{\eta}\right)$$

$$\omega_2 = \omega \cdot N\left(\frac{\ln(b_2 - \theta)}{\eta} - 2\eta\right) + b_2^2 \cdot N\left(-\frac{\ln(b_2 - \theta)}{\eta}\right)$$

where:

- (1) N denotes the cumulative distribution function of the normal distribution;
- (2) $\ln$ denotes the natural logarithm function; and

3. (3) the parameters θ and ω are equal to the following, respectively:

$$\theta = 2 \ln(\mu) - \frac{1}{2} \ln(\omega)$$

and

$$\eta = \sqrt{\ln(\omega) - 2 \ln(\mu)}$$

9.8

Notwithstanding 9.6, where non-proportional reinsurance covers homogeneous risk groups within a segment, the estimated adjustment factor for non-proportional reinsurance must be equal to the following:

$$NP'_h = \frac{\sum_h V_{\text{left}(prem, h)} \cdot NP'_h}{\sum_h V_{\text{left}(prem, h)}}$$

where:

1. (1) $V_{\text{left}(prem, h)}$ denotes the volume measure for premium risk of the homogeneous risk group h determined in accordance with Solvency Capital Requirement - Standard Formula 3A2.3; and
2. (2) NP'_h denotes the estimated adjustment factor for non-proportional reinsurance of homogeneous risk group h determined in accordance with 9.6.

10

Credibility Factor

10.1

1. (1) The credibility factor for segments 1, 5 and 6 set out in Solvency Capital Requirement - Standard Formula 3A3 must be equal to the following:

Time lengths in years | Credibility factor c

---|---

5 | 34%

6 | 43%

7 | 51%

8 | 59%

9 | 67%

10 | 74%

11 | 81%

12 | 87%

13 | 92%

14 | 96%

15 and larger | 100%

1. (2) The credibility factor for:
 1. (a) segments 2 to 4 and 7 to 12 set out in Solvency Capital Requirement - Standard Formula 3A3;
 2. (b) the segments set out in Solvency Capital Requirement - Standard Formula 3C4; and
 3. (c) the revision risk method,
4. must be equal to the following:

Time lengths in years | Credibility factor c

---|---

5 | 34%

6 | 51%

7 | 67%

8 | 81%

9 | 92%

10 and larger | 100%

10.2

For the purposes of 10.1, the time length must be equal to the following:

1. (1) for the premium risk method, the number of accident years for which data are available;
2. (2) for reserve risk method 1, the number of financial years for which data are available;
3. (3) for reserve risk method 2, the number of accident years ___ for which data are available;
4. (4) for the ___revision risk method, the number of financial years for which data are available; and
5. (5) for non-proportional reinsurance method 1 ___ and non-proportional reinsurance method 2, the number of reporting years for which data are available.